Attacks

28 attacks to build, break, and demonstrate. Pick one you actually want to spend three weekends on — the whole point is that you launch it yourself.

> **How this works.** Sign up in pairs, **one person per track**. You each build your own thing and present together as one comparison.
>
> - **Track A — unassisted.** No coding agents, no AI autocomplete. Easier target. Submit a build log.
> - **Track B — agent-assisted.** Any agent, harder target. Submit the full transcript plus a written analysis of everywhere the agent was wrong, stuck, overconfident, or unsafe. *That analysis is the graded artifact.*
> - **Deliverable.** One joint 12-minute video posted 48 hours before your session, plus a 5-minute live Q&A.

> **To claim one:** go to **People** (https://umd.instructure.com/courses/1408572/users) → **Attacks** and join the group corresponding to the attack you want. Two people per group; Ca will not let a third in. Sort out between the two of you who takes Track A and who takes Track

> **Non-negotiable.** Every attack here runs against systems you own or public data you are allowed to use. Read each entry's safety line — it is the boundary of the assignment. If your idea drifts outside it, bring it to me *before* you run it.

All 28 attacks

#	Attack	Date	Effort	Area
01	Make a Bad Certificate Look Good	Tue Sep 15	approachable	Crypto & PKI
02	Factoring the Internet's Weak Keys	Tue Sep 15	moderate	Crypto & PKI
03	Re-identifying an "Anonymized" Dataset	Thu Sep 17	approachable	Privacy
04	Website Fingerprinting on Tor	Tue Sep 22	moderate	Privacy

05	Fingerprinting Obfuscated Proxy Traffic	Tue Sep 22	moderate	Privacy
06	Augur in a Box	Thu Sep 24	moderate	Networks
07	Beating an On-Path Censor	Tue Sep 29	hard	Networks
08	Reading Encrypted Traffic	Tue Sep 29	moderate	Privacy
09	Off-Path TCP Injection	Thu Oct 1	hard	Networks
10	DNS Cache Poisoning	Thu Oct 1	moderate	Networks
11	Congestion Collapse by Optimistic ACK	Thu Oct 1	moderate	Networks
12	The Same Finding, Three Different Answers	Tue Oct 6	approachable	Methodology
13	Geolocation Ground-Truth Audit	Tue Oct 6	moderate	Methodology
14	The Survey Farm	Thu Oct 8	approachable	Methodology
15	Sock-Puppet Steering Audit	Tue Oct 20	moderate	Methodology
16	Live Scam-Infrastructure Hunt	Tue Oct 27	approachable	Crime & scams
17	Follow the Money	Thu Oct 29	approachable	Crime & scams
18	The Scam-Detector Bake-Off	Tue Nov 3	moderate	Crime & scams
19	Has the Copilot Problem Gotten Better?	Thu Nov 5	moderate	AI security
20	How to Prove Any Fuzzer Is Best	Tue Nov 10	moderate	Methodology
21	Control-Flow Hijack Against Modern Defenses	Thu Nov 12	hard	Code & proofs
22	Verified and Still Exploitable	Tue Nov 17	hard	Code & proofs
23	Cheating the Verifier	Thu Nov 19	moderate	Code & proofs
24	Listening to the Sky	Tue Nov 24	approachable	Networks

25	**Jailbreak, Then Fail to Transfer**	Tue Dec 1	hard	AI security
26	**Indirect Injection: Attack vs. Defense**	Tue Dec 1	moderate	AI security
27	**Membership Inference with Canaries**	Thu Dec 3	moderate	AI security
28	**MCP Tool Poisoning**	Thu Dec 3	approachable	AI security

Details

Click any attack to open it.

▶ 01 Make a Bad Certificate Look Good · Tue Sep 15 · approachable

▶ 02 Factoring the Internet's Weak Keys · Tue Sep 15 · moderate

▶ 03 Re-identifying an "Anonymized" Dataset · Thu Sep 17 · approachable

▶ 04 Website Fingerprinting on Tor · Tue Sep 22 · moderate

▶ 05 Fingerprinting Obfuscated Proxy Traffic · Tue Sep 22 · moderate

▶ 06 Augur in a Box · Thu Sep 24 · moderate

▶ 07 Beating an On-Path Censor · Tue Sep 29 · hard

▶ 08 Reading Encrypted Traffic · Tue Sep 29 · moderate

▶ 09 Off-Path TCP Injection · Thu Oct 1 · hard

▼ 10 DNS Cache Poisoning · Thu Oct 1 · moderate

Kaminsky's insight was to stop attacking the record and start attacking the delegation — which gives you unlimited attempts instead of one. The defenses that followed (source-port randomization, 0x20 encoding, cookies) raised the work factor without changing the fundamentals.

Build: your own recursive resolver and your own zone, and poison the cache.

A Classic Kaminsky against a resolver with defenses disabled. Show the race and the win.

B Turn the defenses on and measure the real work factor in 2026 — how many queries, how long, and does any of it survive contact with a modern resolver?

SAFETY. Your resolver, your zone, your network. Never query or poison a resolver you don't own.

Claim this one by joining group **10** under People → Attacks.

▶ 11 Congestion Collapse by Optimistic ACK · Thu Oct 1 · moderate

▶ 12 The Same Finding, Three Different Answers · Tue Oct 6 · approachable

▶ 13 Geolocation Ground-Truth Audit · Tue Oct 6 · moderate

▶ 14 The Survey Farm · Thu Oct 8 · approachable

▶ 15 Sock-Puppet Steering Audit · Tue Oct 20 · moderate

▶ 16 Live Scam-Infrastructure Hunt · Tue Oct 27 · approachable

▶ 17 Follow the Money · Thu Oct 29 · approachable

▶ 18 The Scam-Detector Bake-Off · Tue Nov 3 · moderate

▶ 19 Has the Copilot Problem Gotten Better? · Thu Nov 5 · moderate

▶ 20 How to Prove Any Fuzzer Is Best · Tue Nov 10 · moderate

▶ 21 Control-Flow Hijack Against Modern Defenses · Thu Nov 12 · hard

▶ 22 Verified and Still Exploitable · Tue Nov 17 · hard

▶ 23 Cheating the Verifier · Thu Nov 19 · moderate

▶ 24 Listening to the Sky · Tue Nov 24 · approachable

▶ 25 Jailbreak, Then Fail to Transfer · Tue Dec 1 · hard

▶ 26 Indirect Injection: Attack vs. Defense · Tue Dec 1 · moderate

▶ 27 Membership Inference with Canaries · Thu Dec 3 · moderate

▶ 28 MCP Tool Poisoning · Thu Dec 3 · approachable